

Real-Time Cyberbullying Detection in User Generated Content Using NLP Techniques and SVM

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Abstract—Cyberbullying has become a popular issue in online spaces, seriously affecting people’s mental well-being and overall safety on the digital front. As social media and other platforms keep growing, harmful behaviors are popping up more often, and old-school manual checks just can’t keep up with the volume or the need for instant responses. In this work, we introduce a framework for spotting cyberbullying that’s both explainable and tuned to emotions. It combines machine learning and natural language processing to recognize toxic text. The setup starts with solid text prep: things like normalizing the words, cutting out common stop words, and lemmatizing to get everything in a standard form. Then, it turns the text into features using TF-IDF for term frequency-inverse document frequency. But it doesn’t stop at basic word counts; the model also pulls in deeper context, like the overall sentiment, hints of sarcasm, mentions of gender, and clear signs of personal attacks, all to make predictions more reliable. We train a supervised classifier to separate bullying posts from harmless ones, and it spits out a probability score showing just how damaging a message might be. To fix the black-box problem with most classifiers, our system creates straightforward explanations in plain language, breaking down the words and context that led to each call. On top of that, there’s a part that detects emotions like anger, sadness, or fear, which often show up in this kind of nasty exchange. We built the whole thing into a user-friendly web app with Streamlit, so it handles live inputs from users or crunches through big batches of data at once. When we tested it out, the framework held its own in terms of accuracy, all while staying transparent and easy to use in practice. That makes it a good fit for things like moderating social media, teaching digital safety in schools, or keeping watch on online communities.

Index Terms—Cyberbullying detection, TF-IDF, Support Vector Machine (SVM), Explainable Artificial Intelligence (XAI), Sentiment Analysis, Sarcasm Detection

I. INTRODUCTION

Cyberbullying stands out as one of the biggest types of online harassment. Places like Twitter (which is now X), Facebook, Instagram, TikTok, and Snapchat draw in billions of people, letting users crank out tons of content through comments, photos and videos, and even live broadcasts [1] [2]. What sets it apart from old-school bullying, the kind that happens face-to-face in spots like schools or offices, is how cyberbullying stays hidden, spreads everywhere, and hits right away [4]. Those behind it can hide who they are with fake names, location tags, or VPNs, which makes it way tougher to track them down and hold them responsible.

Cyberbullying hits people no matter their age, gender, or where they live. That being said, teenagers and young adults appear to be particularly vulnerable, as they are still developing their social identities and spend a significant amount of time online. When it happens, victims often face serious mental and social fallout, like dealing with anxiety or depression, pulling away from friends, feeling worse about themselves, thinking about suicide, or even trying it. If it drags on over time, it can lead to lasting problems with schoolwork, relationships, and jobs.

Recent reports really drive home how big this problem is. According to the Cyberbullying Research Center’s 2025 report, more than 37% of kids between 12 and 17 run into abusive content on a weekly basis. A Pew Research Center survey from the same year puts it at about 41% of U.S. teens who’ve dealt with online harassment. At the same time, big tech platforms handle billions of posts every day, which makes

it impossible to count on human moderators alone. On top of that, tougher regulations are coming into play, like the rules for being transparent in the European Union’s Digital Services Act from 2024, and they call for automated moderation that’s clear and accountable. [5]

The usual ways of spotting cyberbullying just don’t cut it for real-time use. Those keyword filters, for one, miss the bigger picture and regularly overlook sarcasm, veiled threats, cultural slang, or those subtle emotional cues. Even standard machine learning setups like Naïve Bayes tend to let too many sneaky or indirect cases slip through, leading to a bunch of false negatives. On top of that, while deep learning models that act like total black boxes can be pretty spot-on, they don’t explain themselves well, which erodes trust from users and makes it hard to meet regulatory standards. To tackle these shortcomings, this paper puts forward a real-time framework for detecting cyberbullying that’s also explainable. It combines natural language processing methods with support vector machine classification [6]. The system relies on organized preprocessing, a mix of feature engineering approaches, and contextual signals to boost its reliability. In particular, it merges TF-IDF for text representation with sentiment polarity ratings, hints of sarcasm, detection of gender references, signs of direct attacks, and emotion identification (such as anger, sadness, fear, or neutral).

The system handles binary classification to spot bullying or non-bullying content, while also producing probabilistic scores that measure the level of risk. What sets it apart from typical methods is its ability to deliver clear, human-interpretable explanations drawn from identified language patterns and emotional cues, which boosts transparency and helps meet regulatory standards [11]. On top of that, it works with both live user inputs in real time and larger batch processes, making it easy to scale up for things like moderating social media, running educational tools, or bolstering online safety setups.

II. LITERATURE SURVEY

Social media’s rapid growth has ramped up cyberbullying cases, turning it into a major issue for both society and technology. What sets cyberbullying apart from old-school bullying is how it thrives in a nonstop, global, and frequently anonymous online space. Researchers have looked into the various behavioral, psychological, and situational elements that fuel this kind of harassment. For instance, Aliyu *et al.* [3] dug into how sharing personal details online can spark cyberbullying, pointing out the privacy risks that come with digital chats. In a related vein, Hu *et al.* [4] investigated links between unwanted online actions and people engaging in cyberbullying, showing just how looped and repeating these abusive behaviors can be. Galyashina and Nikishin [5] took it a step further by examining aggressive info-sharing in terms of protecting users’ core beliefs, and they stressed the wider effects on society from toxic online exchanges.

Back in the early days, automated systems for spotting cyberbullying mostly leaned on classic machine learning methods, paired with manually designed text features. Bayari and Bensefia [6] put together a solid overview of text mining strategies for this purpose, pointing out that tools like Support Vector Machines (SVM), Naïve Bayes, and Decision Trees still hold up well, especially with solid preprocessing and smart feature engineering behind them. Unnava and Parasana [2] ran a direct comparison of various classification algorithms, showing how much the results hinge on the way features are represented. Perdana *et al.* [19] tested out Naïve Bayes, Decision Tree, and K-Nearest Neighbor (KNN) on Instagram comments to catch cyberbullying, and they found that none of these classifiers reliably comes out on top across different sets of data. All this suggests that getting the features just right is key to boosting how well these models perform.

Researchers have increasingly turned to sentiment analysis in cyberbullying detection systems to pick up on the emotional tones hidden in text. For instance, Almufareh *et al.* [1] combined sentiment analysis with machine learning classifiers, which led to better accuracy in spotting bullying. Sharif *et al.* [14] took a closer look at sentiment in individual sentences from cyber trolling tweets, showing just how useful it is to break down polarity at that fine level. Almuayqil *et al.* [16] came up with a framework that boosts sentiment classification by using random minority oversampling to tackle problems with unbalanced data classes. On top of that, Hadi and his group *et al.* [13] used chi-square methods to select key statistical features for classifying cyberbullying based on sentiment, which really highlights why choosing the right text features matters so much.

As deep learning has progressed, transformer-based models have really caught on because of how well they handle context. Ahmed *et al.* [8] looked into these architectures along with ensemble methods for spotting cyberbullying based on traits, and they found they outperformed standard machine learning setups. Teng and Varathan [12] put classical machine learning up against transfer learning options, wrapping up that systems built on BERT deliver better grasp of meaning. Muneer *et al.* [11] took it a step further with a stacking ensemble that uses upgraded BERT features, leading to stronger results in classification. On top of that, hybrid transformer designs have delivered solid outcomes in tasks like detecting misinformation [15], which points to how useful contextual embeddings can be for picking up on tricky language patterns.

Detecting cyberbullying gets tricky because of things like subtle wording, sarcasm, slang, and the way online lingo keeps changing. Parameswaran and colleagues in their work *et al.* [20] pointed out how tough it is to spot the real targets of sarcasm, showing that these kinds of remarks can easily hide someone’s harmful intentions. Maskat and team *et al.* [17] tackled issues with multiple languages by creating an automatic way to label Malay cyberbullying datasets, drawing

on sentiment, emotion, and toxicity levels. Meanwhile, Yi and Zubiaga [18] reviewed approaches to session-based detection, stressing how much conversational flow and sequential patterns matter for making the process more dependable.

Even with all the advances we've seen from deep learning and transfer learning, these techniques still demand a ton of computing power and massive labeled datasets. That makes them tough to use in real-time scenarios or places where resources are tight. On top of that, a lot of those top-notch transformer models aren't easy to interpret, so moderators struggle to figure out why a certain classification happened. Traditional machine learning methods, though, paired with smart feature extraction like TF-IDF and sentiment polarity checks, bring real advantages in terms of speed, how well they scale, and overall clarity.

Therefore, we still need a well-rounded framework for spotting cyberbullying that combines smart preprocessing, solid statistical features, and a sturdy but not-too-heavy classification setup. Drawing from what we've learned in earlier studies, our system uses TF-IDF to represent features, pairs it with sentiment polarity checks, and applies Support Vector Machine (SVM) for classification. The aim here is solid performance without sacrificing clarity, the ability to scale up, or readiness for live use.

III. PROPOSED APPROACH

This paper proposes an explainable and emotion-aware cyberbullying detection framework that integrates TF-IDF features with contextual rule-based indicators. The system performs real-time classification and provides severity scoring, emotion detection, and interpretable explanations for transparent decision-making.

A. System Architecture

The overall architecture is illustrated in Fig. 1. The framework consists of preprocessing, hybrid feature extraction, SVM-based classification, severity estimation, and explanation generation modules.

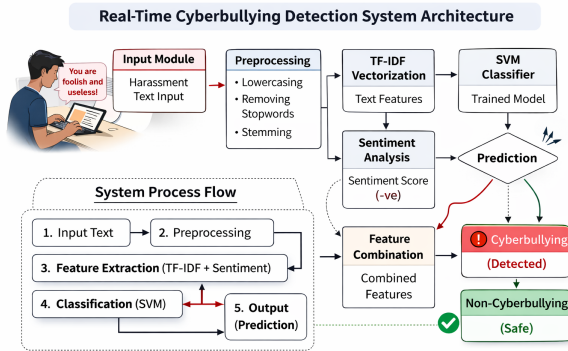


Fig. 1. System Architecture of the Proposed Cyberbullying Detection Framework

Given an input message T_i , the system processes the text

and generates prediction outputs in real time.

B. Text Preprocessing

Let the dataset be defined as $S = \{M_1, M_2, \dots, M_k\}$. Each message undergoes lowercase conversion, URL and special character removal, tokenization, stopword removal, and lemmatization. The cleaned dataset is represented as $S_p = g_{preprocess}(S)$.

C. Hybrid Feature Extraction

To enhance contextual understanding, statistical and rule-based features are combined.

1) TF-IDF Representation

The textual representation is computed as:

$$TFIDF(w, m) = TF(w, m) \times \log \left(\frac{K}{DF(w)} \right)$$

2) Additional Linguistic Indicators

The framework extracts the following contextual features:

- Sentiment polarity $S_i \in [-1, 1]$
- Sarcasm indicator $Sar_i \in \{0, 1\}$
- Gender reference flag $G_i \in \{0, 1\}$
- Direct attack flag $A_i \in \{0, 1\}$

The final hybrid feature vector is:

$$X_i = [TFIDF_i, S_i, Sar_i, G_i, A_i]$$

Table I summarizes the extracted features.

TABLE I
SUMMARY OF EXTRACTED FEATURES

Feature	Description
TF-IDF	Statistical text representation
Sentiment	Polarity score (-1 to 1)
Sarcasm	Rule-based sarcasm detection
Gender Flag	Gender-related reference detection
Attack Flag	Direct abusive phrase detection

D. SVM-Based Classification

A Support Vector Machine (SVM) classifier is employed for binary classification. The SVM determines the optimal separating hyperplane:

$$\theta \cdot z + \beta = 0 \quad (1)$$

The optimization objective is formulated as:

$$\min_{\theta, \beta, \epsilon_i} \frac{1}{2} \|\theta\|^2 + \lambda \sum_{i=1}^m \epsilon_i \quad (2)$$

where θ denotes the weight vector, β the bias term, λ the regularization parameter, and ϵ_i the slack variables.

E. Severity and Explainability

The classifier probability output is transformed into a severity score computed as $Severity_i = P(y_i = 1) \times 100$, representing the risk level (0–100).

Emotion detection categorizes text into Anger, Sadness, Fear, or Neutral using lexicon-based rules. To enhance moderation sensitivity, a safety override assigns Anger if bullying is detected but no explicit emotion is identified.

The explanation module generates interpretable outputs based on detected negative sentiment, sarcasm, gender references, and direct attack indicators, improving transparency over traditional black-box models.

F. Performance Evaluation

The performance of the proposed hybrid cyberbullying detection framework was evaluated using standard classification metrics. Let P_t , N_t , P_f , and N_f denote true positives, true negatives, false positives, and false negatives, respectively.

The evaluation metrics are defined as follows:

$$Accuracy = \frac{P_t + N_t}{P_t + N_t + P_f + N_f}$$

$$precision = \frac{P_t}{P_t + P_f}$$

$$recall = \frac{P_t}{P_t + N_f}$$

$$F1-score = 2 \times \frac{precision \times recall}{precision + recall}$$

The dataset consisted of 8,450 labeled instances, and an 80:20 train-test split was employed, resulting in 1,690 samples for testing.

A baseline model using only TF-IDF features achieved an overall accuracy of **90.12%**. The proposed hybrid SVM model, integrating TF-IDF with sentiment polarity and contextual rule-based indicators, achieved an overall accuracy of **89.35%**.

Although the baseline model attained marginally higher overall accuracy, the proposed model demonstrated improved detection performance for the cyberbullying class (Class 1), achieving higher recall and balanced precision.

TABLE II
PERFORMANCE METRICS OF THE PROPOSED HYBRID SVM MODEL

Model	Accuracy	precision	recall	F1-score
Hybrid SVM	89.35%	0.90	0.92	0.91

The higher recall (92%) for the cyberbullying class indicates improved sensitivity in identifying harmful content, thereby reducing false negatives. In real-world moderation systems, minimizing undetected abusive content is often more critical than marginal gains in overall accuracy.

The confusion matrix of the proposed hybrid SVM classifier is shown in Fig. 2, illustrating the distribution of correct and misclassified instances.

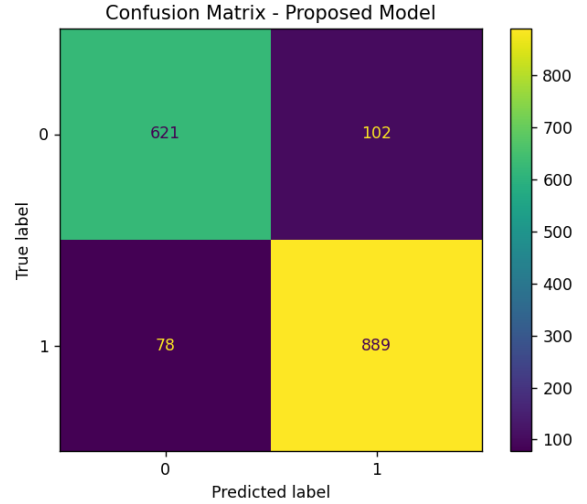


Fig. 2. Confusion Matrix of the Proposed Hybrid SVM Classifier

G. Research Contribution

This work proposes a hybrid and explainable cyberbullying detection framework that integrates statistical TF-IDF representations with contextual rule-based linguistic indicators, including sentiment polarity, sarcasm detection, gender reference flags, and direct attack markers. Unlike conventional approaches that rely solely on statistical text features, the proposed system enhances contextual sensitivity through emotion-aware safeguards and moderation-oriented heuristics. A probability-based severity scoring mechanism is introduced to convert classifier outputs into an interpretable risk scale (0–100), enabling prioritized moderation of high-risk content. Furthermore, an explanation module generates human-readable justifications based on detected linguistic cues, improving transparency compared to traditional black-box classifiers. Experimental evaluation demonstrates strong harmful-content detection capability, achieving a recall of 92% for cyberbullying instances, thereby reducing false negatives and enhancing moderation reliability in real-time social media environments.

IV. RESULTS AND DISCUSSION

The results of the proposed real-time cyberbullying detection framework was evaluated using a supervised learning setup with an 80:20 train-test split. The dataset consisted of 8,450 labeled instances, resulting in 1,690 samples for testing. A Support Vector Machine (SVM) classifier was trained using hybrid features that combined TF-IDF representations with contextual rule-based indicators, including sentiment polarity, sarcasm detection, gender reference flags, and direct attack markers.

To assess the impact of contextual augmentation, the proposed hybrid model was compared against a baseline model utilizing only TF-IDF features. The baseline model achieved an overall accuracy of 90.12%, while the proposed hybrid model achieved an accuracy of 89.35% on the test set.

Although the baseline achieved marginally higher overall accuracy, a class-wise evaluation reveals improved harmful-content sensitivity in the proposed framework. For the cyberbullying category (Class 1), the hybrid model achieved a precision of 0.90, recall of 0.92, and an F1-score of 0.91 across 967 samples. The higher recall indicates improved detection of abusive content, thereby reducing false negatives, which is critical in real-world moderation systems.

For the non-cyberbullying category (Class 0), the model achieved a precision of 0.89, recall of 0.86, and an F1-score of 0.87 across 723 samples. The slightly lower recall for this class indicates that some non-harmful posts were misclassified as abusive, which may result from strong emotional expressions or contextual ambiguity.

The macro-averaged F1-score of 0.89 and weighted average F1-score of 0.89 demonstrate balanced overall performance across both classes. These results indicate that while the hybrid model does not significantly improve raw accuracy over the baseline, it enhances harmful-content detection reliability and provides additional interpretability and severity estimation capabilities, making it more suitable for deployment in safety-sensitive social media environments.

TABLE III
CLASSIFICATION REPORT OF THE PROPOSED HYBRID SVM MODEL

Class	precision	recall	F1-score	support
0 (Non-Cyberbullying)	0.89	0.86	0.87	723
1 (Cyberbullying)	0.90	0.92	0.91	967
Accuracy		0.89		1690
Macro Avg	0.89	0.89	0.89	1690
Weighted Avg	0.89	0.89	0.89	1690

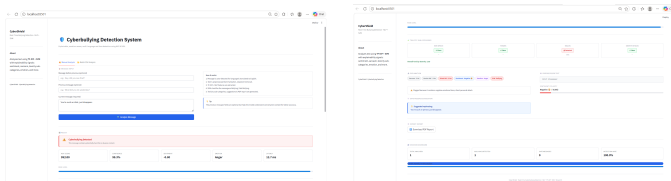


Fig. 3. Output of Cyberbullying Detection System

V. FUTURE WORK

Even though the system we've outlined delivers solid results in classifying content, there are a few ways to build on it and make it more flexible overall. One option is to bring in transformer models like BERT or RoBERTa. These could sharpen the system's grasp of context, especially when it comes to spotting subtle harassment, sarcasm, or how conversations link together. Traditional models that rely on vector spaces often miss those deeper connections, but these embeddings pick up on meaning across full stretches of text, helping to clear up those tricky gray areas.

Another step could involve adapting the setup to handle multiple languages or mixed-language posts. Social media users do that all the time, blending languages in a single

message, so incorporating solid cross-language embeddings could really improve its accuracy in those cases.

Going forward, it might make sense to explore multi-modal methods for spotting cyberbullying, where we pull in text along with images, audio, or videos. So much of the harassment pops up in memes, photos, or short clips that dodge the usual text-only detectors. We should also think about incorporating adaptive learning techniques, since online language evolves all the time. Stuff like periodic retraining or live updates could keep the system sharp against emerging slang or new forms of abusive talk.

Even with the system's solid results, there are still some clear shortcomings. Right now, it only works with text, so it misses any cyberbullying that shows up in images, videos, or audio clips. It relies on basic rules to pick up on sarcasm or those tricky context clues, which can lead to mistakes on more subtle messages. On top of that, the training data sticks to English, so it doesn't handle multilingual content or posts mixing languages very well. And since social media slang and trends change so quickly, the setup could fall behind without regular tweaks or fresh training to catch new forms of abuse or shifts in how people talk online.

Finally, you really need to build in some fairness checks and steps to minimize bias, making sure it handles all kinds of groups and ways of talking equally well. Dealing with bias directly is essential when you're using these moderation tools in sensitive online environments.

REFERENCES

- [1] M. F. Almuftareh, N. Z. Jhanjhi, M. Humayun, G. N. Alwakid, D. Javed, and S. N. Almuayqil, "Integrating sentiment analysis with machine learning for cyberbullying detection on social media," *IEEE Access*, vol. 13, pp. 78348–78359, 2025, doi: 10.1109/ACCESS.2025.3558843.
- [2] S. Unnava and S. R. Parasana, "A study of cyberbullying detection and classification techniques: A machine learning approach," *Engineering, Technology & Applied Science Research*, vol. 14, no. 4, pp. 15607–15613, Aug. 2024, doi: 10.48084/etasr.7621.
- [3] S. Aliyu, K. S. Niksirat, K. Huguenin, and M. Cherubini, "On the role and form of personal information disclosure in cyberbullying incidents," *Proceedings on Privacy Enhancing Technologies*, vol. 2023, no. 4, pp. 468–483, Aug. 2023, doi: 10.56553/popets-2023-0120.
- [4] Y. Hu, E. M. Clancy, and B. Klettke, "Understanding the vicious cycle: Relationships between nonconsensual sexting behaviours and cyberbullying perpetration," *Sexes*, vol. 4, no. 1, pp. 155–166, Feb. 2023, doi: 10.3390/sexes4010013.
- [5] E. I. Galyashina and V. D. Nikishin, "The concepts of aggressive information impact through the lens of internet users' worldview security," *Journal of Siberian Federal University: Humanities & Social Sciences*, vol. 14, no. 11, pp. 1660–1673, Nov. 2021, doi: 10.17516/1997-1370-0848.
- [6] R. Bayari and A. Bensefia, "Text mining techniques for cyberbullying detection: State of the art," *Advances in Science, Technology and Engineering Systems Journal*, vol. 6, no. 1, pp. 783–790, Feb. 2021, doi: 10.25046/aj060187.
- [7] S. Joshi, H. G. Nagariya, N. Dhanotiya, and S. Jain, "Identifying fake profile in online social network: An overview and survey," in *Communications in Computer and Information Science*, vol. 1240, pp. 17–28, 2020, doi: 10.1007/978-981-15-6315-7_2.
- [8] T. Ahmed, S. Ivan, M. Kabir, H. Mahmud, and K. Hasan, "Performance analysis of transformer-based architectures and their ensembles to detect

- trait-based cyberbullying,” *Social Network Analysis and Mining*, vol. 12, no. 1, p. 99, Dec. 2022, doi: 10.1007/s13278-022-00934-4.
- [9] R. Ş. Gün and G. G. Akduman, “What is cyberbullying?” in *Bullying in Media and Beyond*. Turkey: IGI Global, pp. 473–485, doi: 10.4018/978-1-6684-5426-8.CH028.
 - [10] S. Alotaibi and A. Mehmood, “A deep analysis of textual features based cyberbullying detection using machine learning,” in *Proc. IEEE Global Conf. Artificial Intelligence and Internet of Things (GCAIoT)*, Dubai, UAE, Dec. 2022, doi: 10.1109/GCAIoT57150.2022.10019058.
 - [11] A. Muneer, A. Alwadain, M. G. Ragab, and A. Alqushaibi, “Cyberbullying detection on social media using stacking ensemble learning and enhanced BERT,” *Information*, vol. 14, 2023, doi: 10.3390/infoXXXXX.
 - [12] T. H. Teng and K. D. Varathan, “Cyberbullying detection in social networks: A comparison between machine learning and transfer learning approaches,” *IEEE Access*, vol. 11, pp. 55533–55560, 2023, doi: 10.1109/ACCESS.2023.3275130.
 - [13] Z. Hadi, E. Suryadi, A. Akbar, Zaenudin, and R. Muslim, “Cyber bullying sentiment analysis based on social categories using the chi-square test,” *Journal of Computer Technology*, vol. 2, no. 1, pp. 1–9, Jul. 2024.
 - [14] W. Sharif, M. Ashraf, A. Shifa, M. Shahid, Q. Mumtaz, U. Ijaz, M. Anwar, and M. Ikram, “Sentence level sentiment analysis of cyber trolling tweets using machine learning technique,” *Journal of Computational Biomedical Informatics*, vol. 6, no. 2, pp. 1–12, 2024, doi: 10.56979/602/2024.
 - [15] F. Al-Quayed, D. Javed, N. Z. Jhanjhi, M. Humayun, and T. S. Alnusairi, “A hybrid transformer-based model for optimizing fake news detection,” *IEEE Access*, vol. 12, pp. 160822–160834, 2024, doi: 10.1109/ACCESS.2024.3476432.
 - [16] S. N. Almuayqil, M. Humayun, N. Z. Jhanjhi, M. F. Almufareh, and D. Javed, “Framework for improved sentiment analysis via random minority oversampling for user tweet review classification,” *Electronics*, vol. 11, no. 19, p. 3058, Sep. 2022, doi: 10.3390/electronics11193058.
 - [17] R. Maskat, M. F. Zainal, N. Ismail, N. Ardi, A. Ahmad, and N. Daud, “Automatic labelling of Malay cyberbullying Twitter corpus using combinations of sentiment, emotion and toxicity polarities,” in *Proc. 3rd Int. Conf. Algorithms, Computing and Artificial Intelligence*, Dec. 2020, pp. 1–6, doi: 10.1145/3446132.3446412.
 - [18] P. Yi and A. Zubiaga, “Session-based cyberbullying detection in social media: A survey,” *Online Social Networks and Media*, vol. 36, Art. no. 100250, Jul. 2023, doi: 10.1016/j.osnem.2023.100250.
 - [19] R. A. Perdana, C. E. Widodo, and R. Santoso, “Sentiment analysis of Naïve Bayes, decision tree, and K-nearest neighbor (K-NN) algorithms for cyberbullying comments on Instagram accounts,” in *Proc. 11th Int. Conf. Information Technology, Computer, and Electrical Engineering (ICITACEE)*, Aug. 2024, pp. 253–258, doi: 10.1109/ICITACEE62763.2024.10762775.
 - [20] P. Parameswaran, A. Trotman, V. Liesaputra, and D. Evers, “Detecting the target of sarcasm is hard: Really?” *Inf. Process. Manage.*, vol. 58, no. 4, Jul. 2021, Art. no. 102599.